

Module 1: Lesson 4

Introduction to Machine Learning



Outline

- ▶ The setup and data preprocessing
- ▶ Setting up the logistic regression
- ▶ Performance: Confusion matrix and ROC
- ▶ Increasing the complexity of the model

Predicting the probability of default: The setup

The probability of default indicates the likelihood that a borrower will be unable to meet their debt payment obligations (either interest or principal).

- The higher the borrower's probability of default, the higher the interest rate charged by the lender, or the lower the amount the borrower would be eligible to obtain from the bank.

In this lesson, we exploit real-world information on default events from a bank in Israel to train a classification model to predict default cases of individuals.

We will exploit the informativeness that several variables may have in predicting the default behavior of individuals using a logistic regression model.

The model will be of use to banks or credit issuers to estimate the probability of default of an individual who approaches the bank to apply for a loan and who has certain characteristics.

Predicting the probability of default: Data

The data provides information about whether individuals default on their loans ($y = 1$) or not ($y = 0$) and several features (X) that may be informative about the expected default:

- ▶ Age.
- ▶ Education.
- ▶ Years with current employer.
- ▶ Years at current address.
- ▶ Household income.
- ▶ Debt to income ratio.
- ▶ Credit card debt.
- ▶ Other debt.

We transform categorical features (i.e., education) into sets of “dummy” variables that indicate if an individual belongs to a certain category:

- ▶ Use the “dummies” as input features in the ML problem, not the original categorical variable.

Predicting the probability of default: Preprocessing

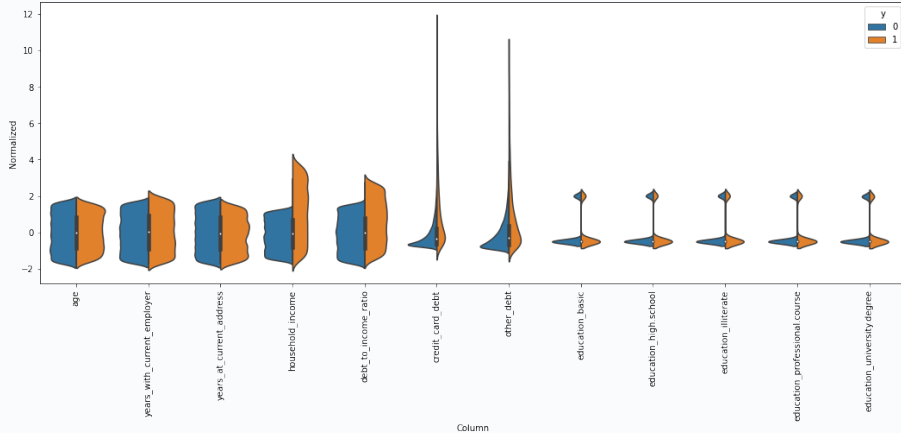
Before training an ML model, we should make sure that we have solid knowledge of the data we are working with.

Having some prior information about the data may very well improve our choices of ML algorithms or hyperparameters and lead to better prediction models.

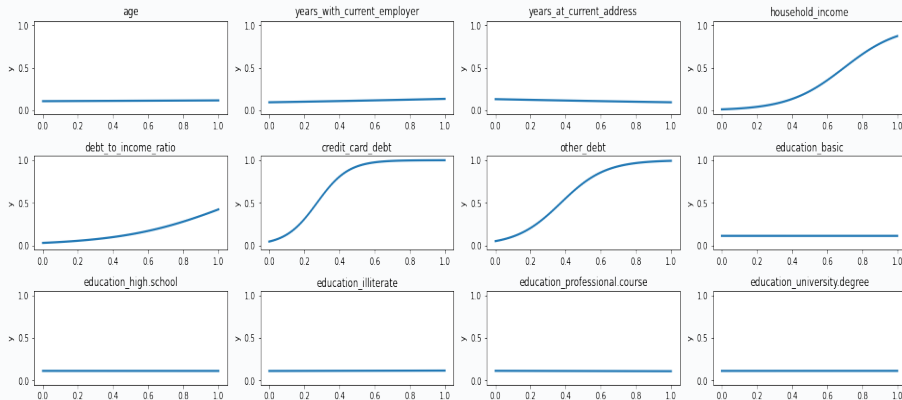
A checklist of steps that we must consider before training an ML model:

1. Analyze the descriptive statistics of the labels and input features.
2. Deal with missing data and outliers.
3. Select the most relevant features.
4. Rescale the features.

Predicting the probability of default: Summary statistics



Predicting the probability of default: Feature selection



Predicting the probability of default: Logistic Regression

We train a logistic regression model to achieve our classification task.

From the previous lesson, recall that the predicted probability of a given instance being a default case is:

$$\hat{p}^{(i)} = \frac{1}{1 + \exp(-x^{(i)}\theta)}$$

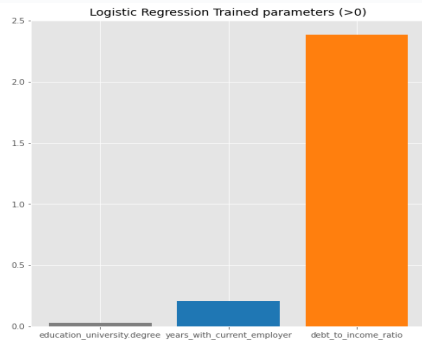
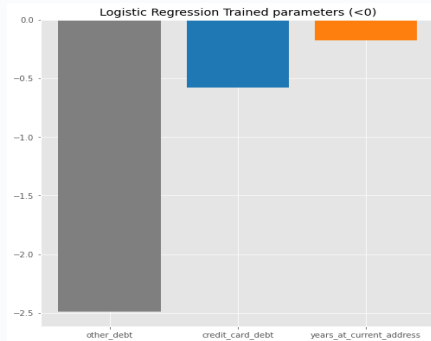
The objective of training is to minimize the cost function:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{p}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]$$

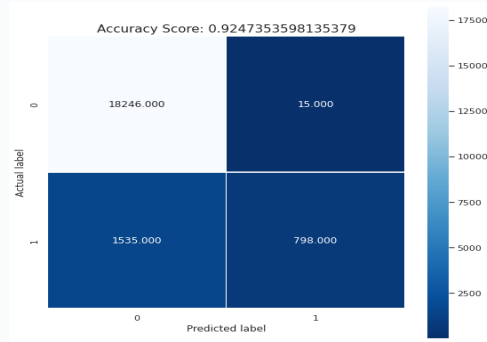
In Python's ScikitLearn, the Logistic Regression model includes by default a Ridge regularization with $\alpha = 1$.

$$\sum_{i=1}^n \theta_i^2$$

Predicting the probability of default: Trained parameters



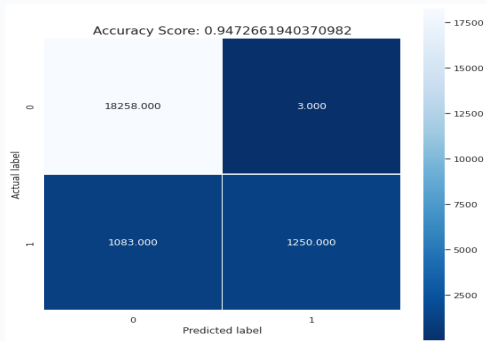
Predicting the probability of default: Confusion matrix



Predicting the probability of default: ROC



Predicting the probability of default: Powers of input features



Summary of Lesson 4

In Lesson 4, we have looked at:

- ▶ The preprocessing tasks that we need to perform before training an ML model
- ▶ Training a Logistic Regression model to predict the probability of default

⇒ **References for this lesson:**

Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 2nd ed., O'Reilly Media, Inc., 2019.

TO DO NEXT: Now, please go to the associated Jupyter Notebook for this lesson to review the full implementation of the default prediction problem.

With this lesson, we end Module 1 as an introduction to Machine Learning. In later modules, you will learn about other useful and more advanced ML methods.